

Quick Math — Kotlin Android Implementation Documentation

Overview

Quick Math is a native Android application built using Kotlin and Jetpack Compose. The app maintains the core functionality of presenting timed math cards during phone sessions while implementing specific trial, OTP, and display logic as per the updated requirements.

1. Onboarding & Trial Logic

The app uses a strict persistence layer (Room/DataStore) to track the user's journey from the first launch.

3-Day Free Trial (No Financials Required)

- * Start Condition: Triggered immediately upon completing the first-time onboarding flow.
- * Trial Properties: - No credit card or bank details required.
 - * hasAnyPremium is set to true for 72 hours.
 - * Strict Mode is fully unlocked during this period.
- * Expiry: After 72 hours, hasAnyPremium reverts to false. Accessing Strict Mode or premium features will then trigger the Subscription/Payment screen.

OTP Verification & Skip Logic

- * Initial State: User is prompted to enter a 10-digit number for OTP verification.
- * Skip for One Day: A "Skip for Now" option is available. If tapped, the app records otpSkippedAt. Verification is suppressed for 24 hours.
- * Mandatory Transition: - OTP becomes mandatory the moment the 3-day Free Trial ends.
 - * If the user skips OTP during the trial, they lose access to premium features immediately after the trial expires until verified.
 - * In mandatory mode, the "Skip" button is removed, and the back button is disabled until a successful login occurs.

2. Core UI & Session Controls

The QM Logo Badge (Top-Left Corner)

- * Interaction: Replaced "Slide" interaction with a Double-Tap gesture.
- * Logic:
 - * Detects two taps within a 320ms window.
 - * Start Session: Changes badge color to Indigo/Green, triggers a spring bounce animation, and starts the background service.
 - * Stop Session: Reverts color, stops timers, and removes any active overlays.
- * Visuals: A gold bubble indicates trial days remaining (e.g., "2d"). A pulsing green dot appears at the bottom-right of the badge when a session is active.

3. Math Card Overlay System

Display Modes (Fixed)

The app utilizes the SYSTEM_ALERT_WINDOW permission to draw over other apps.

1. Half Screen (Bottom Sheet): The card occupies the bottom 40% of the screen. The area above it is dimmed but allows visual context of the background app.
2. Full Screen (Immersive): The card covers 100% of the display. UI elements (Question/Options) scale up (48pt font for questions) to utilize the extra space.

Card Settings

- * Default Duration: 15 seconds.

* Customization: Inside Settings, users can choose: 5s, 10s, 15s, 20s, or 25s.

* Countdown: A circular SVG-style progress ring that transitions from Indigo to Red as the time elapses.

4. Fun Popups & System Permissions

Popups appear between math cards to provide mental breaks.

Popup Logic

* Frequency: Configurable (every 2, 3, or 5 cards).

* Haptics: Continuous vibration pulses every 600ms for the 20-second duration.

* Permission Reference: To ensure these popups appear correctly in the background, the user must enable "Display pop-ups while in background" in Android System Settings (as shown in the provided reference image).

Visual Types

* Type A (Ambient): Staggered scaling blobs with "breathing" animations on a #09090F dark background.

* Type B (Rings): Three concentric arcs spinning at different velocities (1.4s, 2.2s, 3.4s) with a central glowing core.

5. Settings Configuration (Draft/Save Pattern)

Users must tap "Save" to apply changes. Closing without saving triggers a "Discard changes?" confirmation.

Setting

Type

Options / Default

Display Mode

Radio

Half Screen / Full Screen

Card Duration

Picker

5s, 10s, 15s (Default), 20s, 25s

Strict Mode

Toggle

Unlocked in Trial; Paid after

Fun Pop-Ups

Toggle

Enabled/Disabled

Pop-Up Interval

Chip

2, 3, 5 cards

Language

Picker

English, Telugu, Hindi

6. Technical Architecture

* Language: Kotlin 1.9+

* UI Framework: Jetpack Compose (for the main app) + XML Overlay (for the System Alert

Window to ensure stability).

- * Background Work: WorkManager for interval timing and a Foreground Service to keep the session alive during calls.

- * Storage: EncryptedSharedPreferences for the premiumToken and DataStore for general settings.